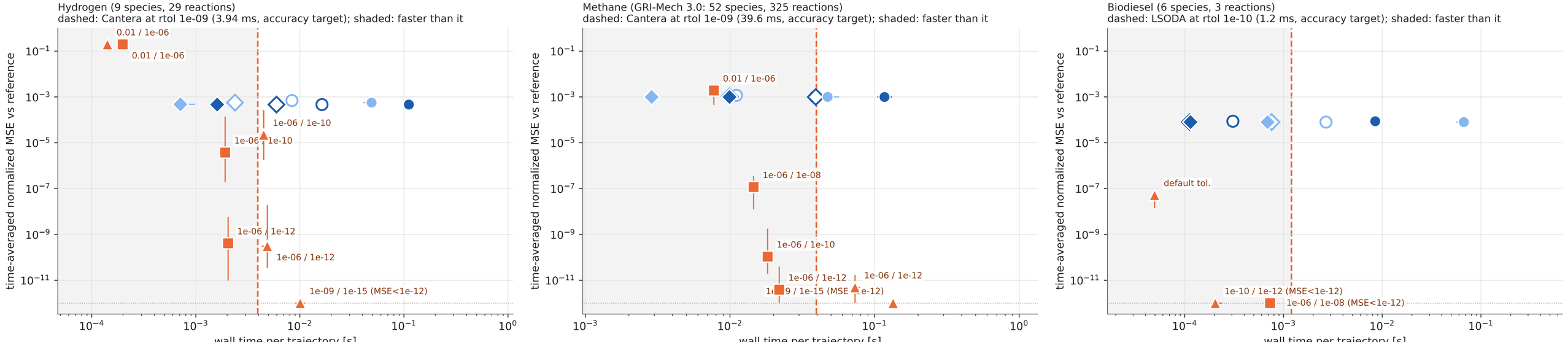
ChemKAN vs detailed-chemistry reference: runtime and accuracy
Hydrogen (9 species, 29 reactions)
dashed: Cantera at rtol 1e-09 (3.94 ms, accuracy target); shaded: faster than it
Methane (GRI-Mech 3.0: 52 species, 325 reactions)
dashed: Cantera at rtol 1e-09 (39.6 ms, accuracy target); shaded: faster than it
Biodiesel (6 species, 3 reactions)
dashed: LSODA at rtol 1e-10 (1.2 ms, accuracy target); shaded: faster than it
time-averaged normalized MSE vs reference
wall time per trajectory [s]
0.01 / 1e-06
1e-06 / 1e-10
1e-06 / 1e-12
1e-09 / 1e-15 (MSE<1e-12)
1e-06 / 1e-08
default tol.
1e-10 / 1e-12 (MSE<1e-12)
Markers: median over conditions x 25 repetitions; bars: IQR. MSE against the reference at its generation tolerance; values below 1e-12 drawn on the dotted floor.
ChemKAN, PyTorch
ChemKAN, Julia port
dark blue: ChemKAN at its training tolerance
light blue: at the other tolerance (H2/CH4 rtol 1e-2, biodiesel 1e-10)
hollow: all conditions in one batched solve
reference, Python (Cantera / LSODA), labels rtol / atol
reference, Julia (Arrhenius.jl / released kinetics)
Python reference at its generation tolerance